

TABLE OF CONTENTS

E-450 AC AND DC DISTRIBUTION SYSTEM

	Page No.
1. SCOPE OF WORK	E-450-2
1.1 General	E-450-2
1.2 AC Distribution System Description.....	E-450-2
1.3 DC Distribution System Description.....	E-450-2
2. CODES AND STANDARDS.....	E-450-2
2.1 Indonesian National Standards	E-450-3
2.2 International Electrotechnical Commission	E-450-3
3. OPERATING AND DESIGN CONDITIONS	E-450-3
4. DETAILED REQUIREMENTS.....	E-450-3
4.1 General	E-450-3
4.2 AC and DC Panel	E-450-3
4.3 Main and Branch Circuits Protective Devices	E-450-4
4.4 Wiring.....	E-450-5
5. PERFORMANCE TESTING AND GUARANTEES	E-450-5
6. QUALITY ASSURANCE/QUALITY CONTROL	E-450-5

E-450 AC AND DC DISTRIBUTION SYSTEM

1. SCOPE OF WORK

1.1 General

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of AC and DC Distribution System for switchyard extension in accordance with Part 4, Subsection 1.2 Summary of Work.

The Contractor shall investigate the possibility to arrange the new or use the existing (if the capacity and reliability sufficient) AC/DC and UPS systems without disturbing the operation of the existing substation.

It is probably possible to use the existing chargers and to integrate the new AC and DC miniature circuit breakers (MCBs) into the existing panels.

Due to the limited space in the AC/DC room, the space has to be saved.

If the above-mentioned arrangement is not possible, the Contractor shall provide a new system.

The technical specification for batteries is covered under Technical Specification E-220.

Redundancy and segregation shall be maintained to provide a highly reliable DC System.

1.2 AC Distribution System Description

The existing AC distribution system is fed from the existing 500 kV substation AC distribution system.

1.3 DC Distribution System Description

The existing DC distribution system consists of 220 volt batteries, chargers, and distribution panels.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4 Subsection 2.5 Codes

and Standards and the following standards:

2.1 Indonesian National Standards

- PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)
SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60269 : Low-voltage fuses
IEC 60947 : Low-voltage switchgear and controlgear
IEC 60529 : Degrees of protection provided by enclosures (IP Code)

3. OPERATING AND DESIGN CONDITIONS

Unless otherwise specified, all equipments for AC and DC distribution system shall be designed for tropical climate along the seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for meteorological, seismic and environmental condition at the site.

4. DETAILED REQUIREMENTS

4.1 General

The Contractor shall design the complete AC and DC distribution system to provide a reliable service.

All low voltage AC (400 volts or below) distribution system components and auxiliaries shall be designed for a minimum 50 kA short circuit rating.

All 220 volt DC distribution system component and auxiliaries shall be designed for a minimum 25 kA short circuit rating. All other DC system component shall be designed to withstand the available short circuit current.

The Contractor shall submit to the Owner the detailed design of AC and DC Distribution System for review.

4.2 AC and DC Panel

The existing AC/DC panels where possible may be used. In case of new panel requirement, the following specification shall be applied:

- (1) All panels shall be designed and fabricated in accordance with the latest applicable standards and Technical Specification E-280 Panel Design.
- (2) Each panel shall be enclosed in a cabinet. Indoor panel board shall be mounted in dust resisting cabinets with a full gasket between the boxes and the cabinet front and between the cabinet door and the cabinet front. Outdoor panel board shall be mounted in weatherproof cabinets rated IPW 55 as per IEC 60529.
- (3) Each cabinet shall be code gage steel and shall consist of a box with removable front complete with hinged door, latch and master keying lock.
- (4) The box shall be flanged galvanized sheet steel and fabricated with straight edges and square corners.
- (5) The panel board interior shall be equipped with a dead-front shield separate in from the cabinet front.
- (6) A circuit directory card holder and card with a clear plastic covering shall be provided on the inside of each cabinet door.
- (7) Rigid buses of copper or copper alloy shall be installed to provide consecutive phasing of branch circuit connections.
- (8) Solid neutral bus in AC panel boards shall have solderless connectors, numbered to agree with branch circuits. The bus shall be insulated from the cabinet and shall be independently connected to grounding system.
- (9) Equipment grounding bus in AC panel with main lug shall have solder less connectors, numbered to agree with branch circuits. Bus shall be sized in accordance with Technical Specification E-001.

4.3 Main and Branch Circuits Protective Devices

The voltage rating and the short circuit rating of the protective over current devices for main and branch circuits shall be as specified in Specification E-001.

Main circuit protective devices shall be triple-pole (AC) or double pole (DC) thermal magnetic type circuit breakers, designed for automatic tripping and shall indicate “ON”, “OFF” and “TRIP” lettering in front. The trip range of the main circuit breaker shall be coordinated with the associated branch circuit breakers and expected load.

Triple-pole, double-pole and single-pole thermal magnetic type branch circuit breakers shall be provided and shall be designed for automatic tripping.

Branch circuit breakers installed in the panel boards shall be identified with odd numbers on the left and even numbers on the right.

Circuit breakers shall be designed and operated at the ambient temperature of 40 °C.

4.4 Wiring

The wiring shall be as specified in Technical specification E-280.

5. PERFORMANCE TESTING AND GUARANTEES

Factory Tests shall be performed and certified test reports shall be submitted to the Owner to confirm the compliance with all technical specifications. All tests shall be in accordance with the applicable IEC standards.

IEC standard. The type test certificates shall be subject to the review and approval of the Owner.

6. QUALITY ASSURANCE/QUALITY CONTROL

The quality assurance/quality control of AC and DC distribution system shall conform to the requirements of Part 4, Section 2, Subsection 2.7. The procedures as specified in, Subsection 2.7 shall include, but not be limited to, the followings:

- (1) Continuity and Wiring Tests.
- (2) Insulating and High Voltage Tests.
- (3) Packaging and Shipping.
- (4) Welding.
- (5) Cleaning and Painting.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Shop Test Reports.
- (2) Inspection Reports.
- (3) All other documentation required by applicable codes and standards.

Last page of E-450)